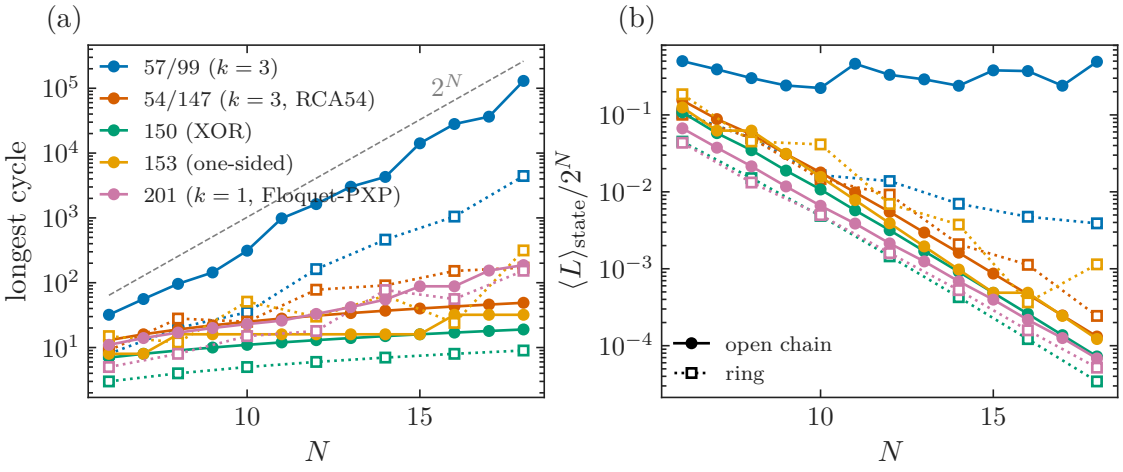

(a)
(b)
longest cycle
$\langle L \rangle_{\text{state}}/2^N$
N
N
57/99 ($k = 3$)
54/147 ($k = 3$, RCA54)
150 (XOR)
153 (one-sided)
201 ($k = 1$, Floquet-PXP)
2^N
open chain
ring
10^1
10^2
10^3
10^4
10^5
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10
15